

CHERNYSHEVSKY, Russian Federation

WMO: 247240

Lat: 63.0171N

Lon: 112.4340E

Elev: 422

StdP: 96.36

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-44.7	-41.9	-46.8	0.0	-44.7	-44.3	0.1	-41.9	9.4	-21.7	8.9	-24.4	3.5	230	0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	27.8	17.8	25.7	16.7	23.7	15.7	18.7	25.7	17.5	23.8	16.5	22.4	3.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	12.1	21.5	15.0	11.2	20.1	13.9	10.4	19.2	55.0	25.8	51.2	24.0	48.0	22.5	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.9	-44.7	31.1	3.4	1.3	-47.2	32.1	-49.2	32.9	-51.1	33.6	-53.6	34.6		
(5)			-44.7	20.5	3.3	1.3	-47.1	21.4	-49.0	22.2	-50.9	22.9	-53.3	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.2	-29.5	-26.3	-15.0	-4.5	4.3	14.8	17.6	13.7	5.1	-6.2	-21.3	-28.9
	DBStd	18.02	7.73	7.26	7.42	5.73	5.27	5.33	4.39	4.04	4.67	6.00	8.33	8.06
	HDD10.0	6483	1226	1015	776	434	191	18	2	13	159	503	939	1206
	HDD18.3	9047	1484	1248	1034	684	436	130	67	150	398	762	1189	1464
	CDD10.0	553	0	0	0	0	13	161	238	129	11	0	0	0
	CDD18.3	75	0	0	0	0	0	23	44	7	0	0	0	0
	CDH23.3	491	0	0	0	0	1	156	293	42	0	0	0	0
	CDH26.7	104	0	0	0	0	0	31	67	7	0	0	0	0
Wind	WSAvg	3.5	4.0	3.5	3.6	3.7	3.6	3.1	2.8	3.0	3.4	3.8	3.7	3.8
Precipitation	PrecAvg	391	15	13	17	22	33	55	42	54	47	48	28	17
	PrecMax	635	34	26	26	37	63	114	108	147	102	90	45	30
	PrecMin	230	7	5	4	4	13	18	7	7	8	15	13	5
	PrecStd	81	6	5	7	10	13	27	22	28	24	20	10	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.6	-5.5	4.1	10.7	21.2	29.9	30.9	27.7	19.8	7.2	-0.9	-5.5
		MCWB	-10.1	-6.0	0.0	4.1	11.1	17.8	19.9	17.6	12.5	4.1	-1.7	-5.9
	2%	DB	-13.5	-9.8	-0.2	8.3	17.9	27.0	28.6	24.2	16.2	5.0	-2.7	-10.7
		MCWB	-14.0	-10.4	-2.7	2.6	9.6	16.7	18.5	16.0	10.2	2.9	-3.3	-11.1
	5%	DB	-15.9	-12.6	-2.7	5.9	15.0	24.9	26.5	22.1	13.6	2.8	-5.3	-14.7
		MCWB	-16.3	-13.1	-4.7	1.1	7.8	15.7	17.3	15.1	8.9	1.0	-5.9	-15.1
	10%	DB	-19.2	-15.7	-5.0	3.6	12.4	22.9	24.6	20.1	11.4	1.1	-9.0	-17.7
		MCWB	-19.5	-16.1	-6.5	-0.3	6.1	14.7	16.5	14.0	7.9	-0.2	-9.4	-18.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.0	-6.0	0.2	4.3	11.9	19.0	21.0	18.6	12.9	4.7	-1.3	-5.8
		MCDB	-9.8	-5.4	3.4	10.2	19.6	27.6	29.5	25.5	18.6	6.6	-0.9	-5.4
	2%	WB	-13.9	-10.3	-2.5	2.9	10.0	17.6	19.5	17.0	11.2	3.1	-3.3	-11.0
		MCDB	-13.4	-9.8	-0.1	7.8	17.1	25.2	26.6	22.2	14.8	4.9	-2.7	-10.6
	5%	WB	-16.3	-13.1	-4.5	1.4	8.2	16.5	18.3	15.8	9.7	1.4	-5.9	-15.0
		MCDB	-16.0	-12.7	-2.8	5.1	14.5	23.5	24.6	20.5	12.5	2.5	-5.3	-14.6
	10%	WB	-19.5	-16.1	-6.4	0.1	6.6	15.3	17.2	14.7	8.3	-0.1	-9.5	-18.1
		MCDB	-19.2	-15.8	-5.0	3.2	11.8	21.6	23.1	19.0	11.1	0.8	-9.0	-17.8
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.8	7.5	7.6	7.6	8.7	8.1	7.3	5.7	4.5	5.3	5.5
		MCDBR	7.8	8.1	7.8	8.4	10.8	10.1	10.0	9.9	8.6	5.4	7.7	8.2
		MCWBR	7.7	8.0	6.4	5.5	6.3	4.7	4.5	4.6	4.7	4.3	7.7	8.2
	5% WB	MCDBR	7.8	8.1	7.8	7.9	10.5	9.6	9.3	8.8	7.6	4.9	7.9	8.1
		MCWBR	7.8	7.9	6.5	5.5	6.4	4.7	4.7	4.7	5.0	4.4	7.8	8.0
Clear-Sky Solar Irradiance	taub	0.210	0.256	0.289	0.320	0.348	0.388	0.430	0.419	0.314	0.271	0.204	0.186	
	taud	1.918	2.056	2.084	2.135	2.328	2.342	2.207	2.181	2.492	2.298	1.942	1.872	
	Ebn at Noon	404	674	799	856	866	832	777	743	793	670	404	220	
	Edh at Noon	34	69	101	120	109	110	123	116	69	54	32	17	
All-Sky Solar Radiation	RadAvg	0.28	1.20	2.92	4.92	5.45	5.73	5.21	3.90	2.34	1.05	0.37	0.12	
	RadStd	0.01	0.07	0.12	0.26	0.34	0.43	0.34	0.27	0.23	0.09	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+1.25	N/A	N/A	+2.00	+0.47	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	+1.25	N/A	N/A	+2.00	+0.47	N/A	N/A	N/A	N/A

Nomenclature: See separate page